# Classification Techniques in Machine Learning

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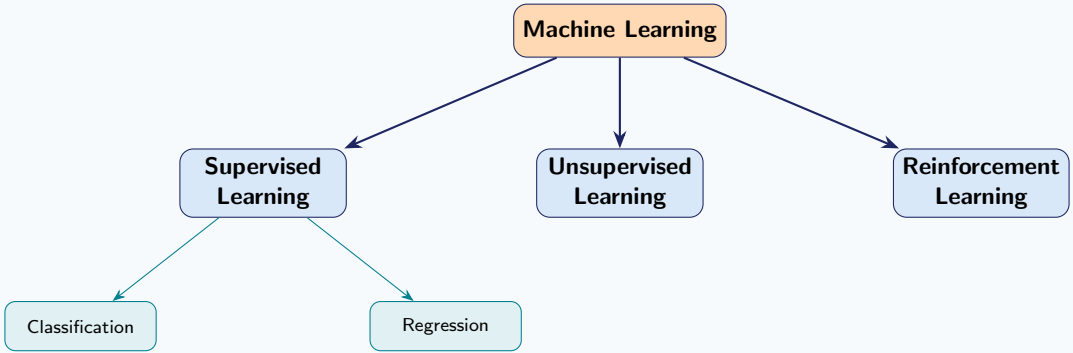
# Outline

# What is Machine Learning?

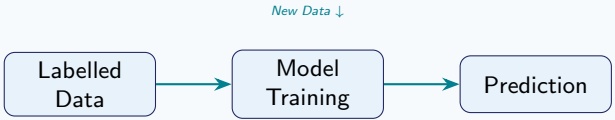
Machine learning enables systems to learn from data and improve without being explicitly programmed.



**Definition**

A model **learns from past labelled data** and makes predictions on new, unseen inputs.

**Pipeline:**




**Types of Supervised Learning:**

- **Classification** — predict a discrete label
- **Regression** — predict a continuous value

**Covered in this lecture:**

1. Logistic Regression
2. Support Vector Machine (SVM)
3. K-Nearest Neighbour (KNN)
4. Naïve Bayes
5. Decision Trees

# What is Logistic Regression?

**Definition**

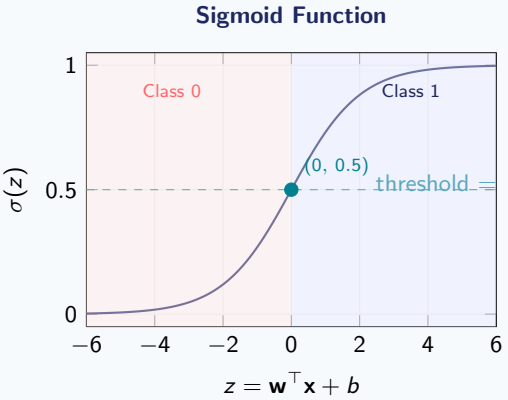
Logistic Regression is a **supervised classification** algorithm that models the probability that an input **x** belongs to a given class using the **sigmoid function**.

**Key :** instead of predicting a value directly, predict

$$P(y = 1 \mid \mathbf{x}) = \sigma(z), \qquad z = \mathbf{w}^\top \mathbf{x} + b$$

**Sigmoid (Logistic) Function**

$$\sigma(z) = \frac{1}{1 + e^{-z}} \in (0, 1)$$




**Decision rule:**

$$\hat{y} = \begin{cases} 1 & \sigma(z) \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

$\sigma(z) \to 0$ as $z \to -\infty$; $\sigma(z) \to 1$ as $z \to +\infty$

### Log-Loss (Binary Cross-Entropy)

$$J(\mathbf{w}, b) = -\frac{1}{n}\sum_{i=1}^{n}\Big[y_i \log \hat{p}_i + (1 - y_i)\log(1 - \hat{p}_i)\Big]$$

where $\hat{p}_i = \sigma(\mathbf{w}^\top \mathbf{x}_i + b)$.

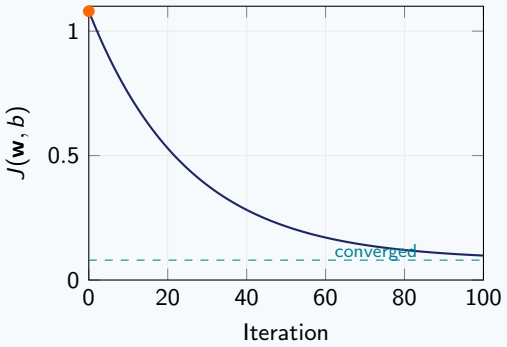
**Why not MSE?** Log-loss is convex — guaranteed global minimum.

### Gradient Descent Update

$$\mathbf{w} \leftarrow \mathbf{w} - \alpha \frac{\partial J}{\partial \mathbf{w}}, \qquad b \leftarrow b - \alpha \frac{\partial J}{\partial b}$$

$$\frac{\partial J}{\partial \mathbf{w}} = \frac{1}{n}\sum_{i=1}^{n}(\hat{p}_i - y_i)\,\mathbf{x}_i$$

**Log-Loss Convergence**



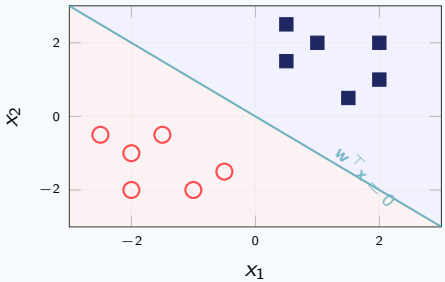

**Regularisation** (prevent overfitting):

$$J_{\text{reg}} = J + \frac{\lambda}{2n}\|\mathbf{w}\|^2 \quad \text{(L2 / Ridge)}$$

**Linear Decision Boundary:** The boundary is where $\sigma(z) = 0.5$, i.e. $z = 0$:

$$\mathbf{w}^\top \mathbf{x} + b = 0$$

**Decision Boundary**




**Multi-class: Softmax Regression**

For $K > 2$ classes, generalise to **softmax**:

$$P(y = k \mid \mathbf{x}) = \frac{e^{\mathbf{w}_k^\top \mathbf{x} + b_k}}{\sum_{j=1}^{K} e^{\mathbf{w}_j^\top \mathbf{x} + b_j}}$$

One-vs-Rest (OvR): train $K$ binary classifiers.

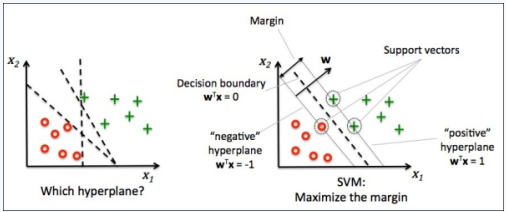
**Key Properties**

- **Output:** probability $\in (0, 1)$ — well-calibrated
- **Boundary:** linear in feature space
- **Convex loss** — efficient, guaranteed convergence
- **Sensitive** to outliers and feature , normalise inputs
- **Regularise** with L1 (sparse) or L2 (small weights)

**Definition**

An **SVM** is a supervised ML algorithm for classification *or* regression. It finds the **hyperplane** that best separates the classes with maximum margin.

**SVM Workflow:**




**Key Concepts:**

- **Hyperplane** — decision boundary
- **Support Vectors** — nearest data points to the boundary
- **Margin** — distance from boundary to support vectors
- Goal: maximise the margin

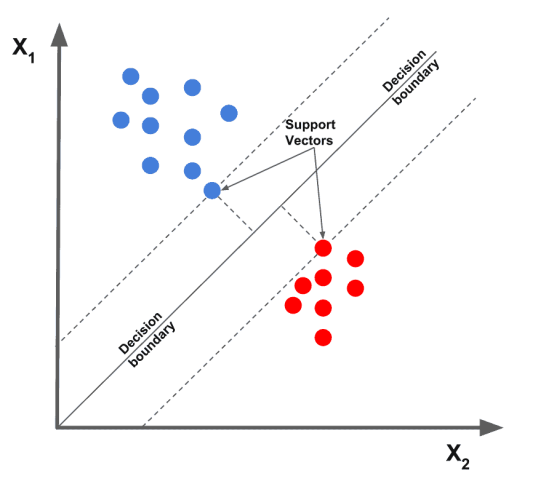
# SVM — Sample Dataset

Height-Weight data used to illustrate classification:

| Female | |
|---|---|
| **Height (cm)** | **Weight (kg)** |
| 174 | 65 |
| 174 | 88 |
| 175 | 75 |
| 180 | 65 |
| 185 | 80 |

| Male | |
|---|---|
| **Height (cm)** | **Weight (kg)** |
| 179 | 90 |
| 180 | 80 |
| 183 | 80 |
| 187 | 85 |
| 182 | 72 |

Goal: find a hyperplane that separates Female from Male data points.



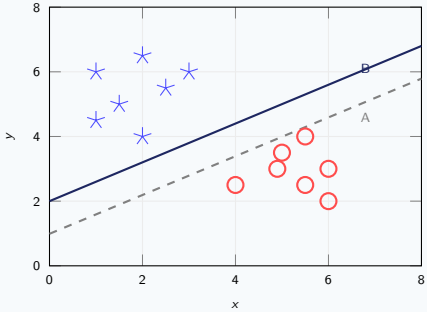

**Optimal Hyperplane**

The hyperplane that **maximises the margin** between the two classes.

- Dashed lines = margin boundaries
- Points on dashed lines = **support vectors**
- The distance between support vectors and hyperplane should be as large as possible

# SVM — Identifying the Right Hyperplane

**Scenario 3: Hyperplane A vs B**




**Scenario 3**

Both A and B separate the classes correctly. **B has a higher margin** → B is the correct choice.

**Scenario 4: Outlier**

When an outlier exists from one class, SVM uses a **soft-margin** formulation to ignore it.

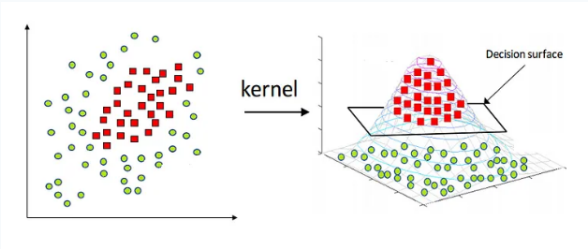
**Scenario 5: Non-linear**

Classes not linearly separable → use the kernel trick to add features.

**Problem:** Red circles and blue stars cannot be separated by a line in $(x, y)$.
**Solution:** Introduce a new feature

$$z = x^2 + y^2$$

- $z \geq 0$ always (squared sum)
- Red circles: close to origin $\Rightarrow$ low $z$
- Blue stars: away from origin $\Rightarrow$ high $z$




Projecting to $(x, z)$ makes the classes linearly separable.

# K-Nearest Neighbour — Basic Idea

**Intuition**

*"If it walks like a duck, quacks like a duck, then it's probably a duck."*

**$k$-NN Classification Rule**

Assign to a test sample the **majority class label** of its $k$ nearest training samples.

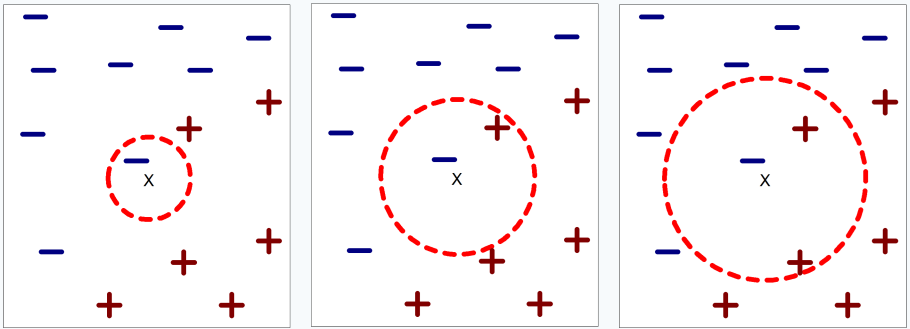
**Process:**


**Key notes:**

- $k$ usually chosen **odd** (avoid ties)
- $k = 1$: nearest-neighbour rule
- **Rule of thumb:** $k \approx \sqrt{N}$

**Definition**

The $k$-**nearest neighbours** of a record **x** are the $k$ data points in the training set with the smallest distance to **x**.




(a) 1-nearest neighbor (b) 2-nearest neighbor (c) 3-nearest neighbor

**Basic $k$-NN** (majority vote):

$$\hat{f}(q) = \arg\max_{v \in V} \sum_{i=1}^{k} \delta\big(v,\ f(x_i)\big)$$

**Distance-Weighted $k$-NN** (closer $\Rightarrow$ more influence):

$$\hat{f}(q) = \arg\max_{v \in V} \sum_{i=1}^{k} \frac{1}{\big[d(x_i, x_q)\big]^2}\, \delta\big(v,\ f(x_i)\big)$$

General kernel functions (e.g. *Parzen Windows*) may replace inverse distance.

**Predicting Continuous Values:** Replace majority vote with weighted average:

$$\hat{f}(q) = \frac{\sum_{i=1}^{k} w_i\, f(x_i)}{\sum_{i=1}^{k} w_i} \qquad \text{(unweighted: } w_i = 1\text{)}$$

## Issues in $k$-NN

- Choice of $k$
- Distance metric
- Computational complexity
- Size of training set
- Dimensionality of data

## Choosing $k$

- $k$ too **small** $\Rightarrow$ sensitive to noise
- $k$ too **large** $\Rightarrow$ neighbourhood includes other classes

**Rule of Thumb:**

$$k \approx \sqrt{N}$$

where $N$ = number of training points.

| Metric | Formula |
|---|---|
| **Minkowski** | $D = \left(\sum \lvert x_i - y_i \rvert^r\right)^{1/r}$ |
| **Euclidean** | $D = \sqrt{\sum (x_i - y_i)^2}$ |
| **Manhattan** | $D = \sum \lvert x_i - y_i \rvert$ |
| **Canberra** | $D = \sum \dfrac{\lvert x_i - y_i \rvert}{\lvert x_i + y_i \rvert}$ |
| **Chebyshev** | $D = \max \lvert x_i - y_i \rvert$ |
| **Quadratic** | $D = (\mathbf{x} - \mathbf{y})^T Q (\mathbf{x} - \mathbf{y})$, $Q$: pos.-def. weight matrix |
| **Mahalanobis** | $D = [\det V]^{1/m} (\mathbf{x} - \mathbf{y})^T V^{-1} (\mathbf{x} - \mathbf{y})$ |
| **Correlation** | $D = \dfrac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$ |
| **Chi-square** | $D = \sum \dfrac{1}{\text{sum}_i} \left( \dfrac{x_i}{\text{size}_x} - \dfrac{y_i}{\text{size}_y} \right)^2$ |
| **Kendall** | $D = 1 - \dfrac{2}{n(n-1)} \sum_{i>j} \text{sign}(x_i - x_j) \text{sign}(y_i - y_j)$ |

# Bayes' Theorem — Foundation

## General Form

For events $A, B$ with $P(B) > 0$:

$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)}$$

**In classification** ($\mathbf{X}$=$(X_1, \ldots, X_n)$, class $C$):

$$P(C \mid \mathbf{X}) = \frac{P(\mathbf{X} \mid C)\,P(C)}{P(\mathbf{X})} \;\propto\; P(\mathbf{X} \mid C)\,P(C)$$

## Term-by-Term

| | |
|---|---|
| $P(C \mid \mathbf{X})$ | **Posterior**: class given data |
| $P(\mathbf{X} \mid C)$ | **Likelihood**: data given class |
| $P(C)$ | **Prior**: base-rate of class |
| $P(\mathbf{X})$ | **Evidence**: constant normaliser |

## Key Insight

$P(\mathbf{X})$ is the **same for every class**, so it cannot change which class wins.

We drop it and compare just:

$$\text{score}(c) = P(\mathbf{X} \mid c) \cdot P(c)$$

## MAP Decision Rule

Predict the class with the highest score:

$$c^* = \arg\max_{c \in \mathcal{C}} \; P(\mathbf{X} \mid c)\,P(c)$$

# From Bayes to Naïve Bayes — The Key Assumption

## Why the Full Joint is Intractable

Exact Bayes needs $P(X_1, \ldots, X_n \mid C)$ — the *full joint*. With $n$ binary features: $\mathbf{2^n}$ probabilities to estimate.
$n$=30 $\Rightarrow$ **1 billion** parameters. Completely impractical!

**Naïve Bayes Assumption** — features are conditionally independent given $C$:

$$P(X_1, \ldots, X_n \mid C) = \prod_{i=1}^{n} P(X_i \mid C) = P(X_1 \mid C) \cdot P(X_2 \mid C) \cdots P(X_n \mid C)$$

## Why This Helps

Parameters drop from $\mathbf{2^n}$ to $\mathbf{n} \times |\mathcal{C}|$.

$n$=30, 2 classes $\Rightarrow$ **60 estimates** vs. $10^9$.

Each $P(X_i \mid C)$ learned independently from data.

## Why It Is "Naïve"

Real features are **rarely independent** (e.g. *word frequency* and *document length*).

Yet Naïve Bayes **works surprisingly well** in spam detection, text classification, and medical diagnosis.

# Naïve Bayes — Algorithm

## Learning Phase

1: **for all** target $c_i \in \{c_1, \ldots, c_L\}$ **do**
2: $\quad \hat{P}(C{=}c_i) \leftarrow$ estimate from $S$
3: $\quad$ **for all** attr. value $a_{jk}$ of $x_j$ **do**
4: $\quad\quad \hat{P}(X_j{=}a_{jk} | C{=}c_i) \leftarrow$ est.
5: $\quad$ **end for**
6: **end for**

Output: conditional probability tables

## Test Phase

**Require:** Instance $X' = (a'_1, \ldots, a'_n)$

1: Look up tables
2: Assign $c^*$ to $X'$ if

$$\prod_i \hat{P}(a'_i | c^*)\, \hat{P}(c^*) > \prod_i \hat{P}(a'_i | c)\, \hat{P}(c)$$

for all $c \neq c^*$.

# Naïve Bayes — Example: Play Tennis (Training Data)

| Day | Outlook | Temp | Humidity | Wind | PlayTennis |
|---|---|---|---|---|---|
| D1 | Sunny | Hot | High | Weak | No |
| D2 | Sunny | Hot | High | Strong | No |
| D3 | Overcast | Hot | High | Weak | Yes |
| D4 | Rain | Mild | High | Weak | Yes |
| D5 | Rain | Cool | Normal | Weak | Yes |
| D6 | Rain | Cool | Normal | Strong | No |
| D7 | Overcast | Cool | Normal | Strong | Yes |
| D8 | Sunny | Mild | High | Weak | No |
| D9 | Sunny | Cool | Normal | Weak | Yes |
| D10 | Rain | Mild | Normal | Weak | Yes |
| D11 | Sunny | Mild | Normal | Strong | Yes |
| D12 | Overcast | Mild | High | Strong | Yes |
| D13 | Overcast | Hot | Normal | Weak | Yes |
| D14 | Rain | Mild | High | Strong | No |

$$P(\text{Play=Yes}) = \frac{9}{14}, \qquad P(\text{Play=No}) = \frac{5}{14}$$

# Naïve Bayes — Conditional Probability Tables

| **Outlook** | Yes | No |
|---|---|---|
| Sunny | 2/9 | 3/5 |
| Overcast | 4/9 | 0/5 |
| Rain | 3/9 | 2/5 |

| **Temperature** | Yes | No |
|---|---|---|
| Hot | 2/9 | 2/5 |
| Mild | 4/9 | 2/5 |
| Cool | 3/9 | 1/5 |

| **Humidity** | Yes | No |
|---|---|---|
| High | 3/9 | 4/5 |
| Normal | 6/9 | 1/5 |

| **Wind** | Yes | No |
|---|---|---|
| Strong | 3/9 | 3/5 |
| Weak | 6/9 | 2/5 |

# Naïve Bayes — Test Phase Example

**New instance:** $x' = (\text{Sunny, Cool, High, Strong})$

**Play = Yes:**

$$P(\text{Sunny}|\text{Yes}) = 2/9$$
$$P(\text{Cool}|\text{Yes}) = 3/9$$
$$P(\text{High}|\text{Yes}) = 3/9$$
$$P(\text{Strong}|\text{Yes}) = 3/9$$
$$P(\text{Yes}) = 9/14$$
$$P(\text{Yes}|x') \approx 0.0053$$

**Play = No:**

$$P(\text{Sunny}|\text{No}) = 3/5$$
$$P(\text{Cool}|\text{No}) = 1/5$$
$$P(\text{High}|\text{No}) = 4/5$$
$$P(\text{Strong}|\text{No}) = 3/5$$
$$P(\text{No}) = 5/14$$
$$P(\text{No}|x') \approx 0.0206$$

**Decision**

Since $P(\text{Yes}|x') < P(\text{No}|x')$, we label $x'$ as **No** (Do not play tennis).

# Decision Trees — Outline

**Topics covered:**

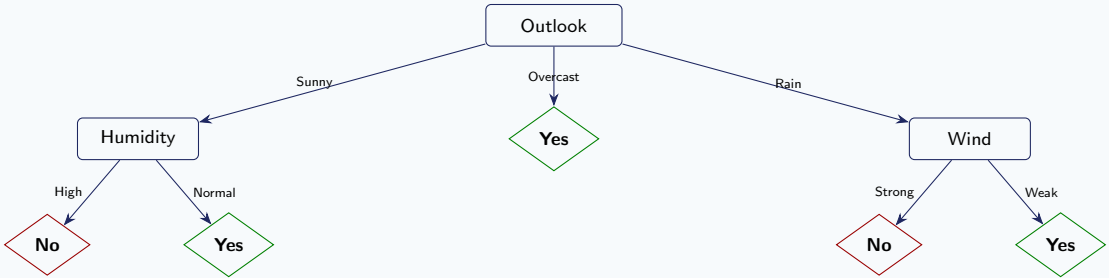
1. Decision Tree Representations
2. ID3 and C4.5 algorithms (Quinlan 1986)
3. CART algorithm (Breiman et al. 1985)
4. Entropy & Information Gain
5. Overfitting

**Key idea**

A tree of **if-then rules** that partitions the feature space into axis-parallel rectangles, each labelled with a class.

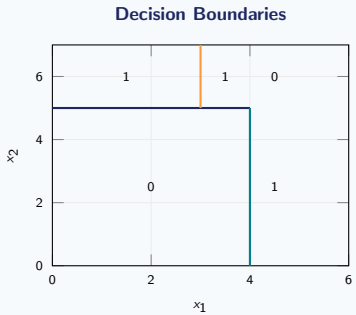
- **Internal nodes** — test a feature $x_j$
- **Branches** — feature values
- **Leaf nodes** — class label $h(\mathbf{x})$

# Decision Trees — Example Tree (Play Tennis)

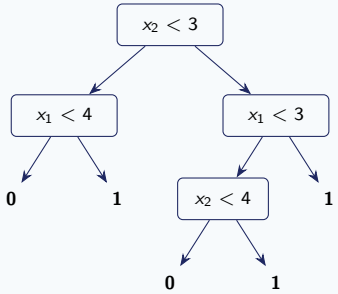



Example: $\mathbf{x} = (\textit{Sunny}, \textit{Hot}, \textit{High}, \textit{Strong})$ → follows Outlook→Sunny → Humidity→High → No. Temperature is irrelevant.

**Decision boundaries** are axis-parallel rectangles:




**Corresponding tree:**



# Decision Trees — Learning Algorithm (GrowTree)

Given $S = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\}$, $\mathbf{x} = (x_1, \ldots, x_d)$, $x_j, y \in \{0, 1\}$:

**GrowTree($S$)**

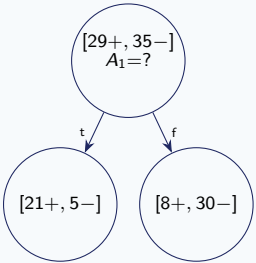
1: **if** $y = 0$ for all $\langle \mathbf{x}, y \rangle \in S$ **then return** new leaf(0)
2: **else if** $y = 1$ for all $\langle \mathbf{x}, y \rangle \in S$ **then return** new leaf(1)
3: **else**
4: choose best attribute $x_j$
5: $S_0 \leftarrow$ all $\langle \mathbf{x}, y \rangle \in S$ with $x_j = 0$
6: $S_1 \leftarrow$ all $\langle \mathbf{x}, y \rangle \in S$ with $x_j = 1$ **return** new node($x_j$, GrowTree($S_0$), GrowTree($S_1$))
7: **end if**

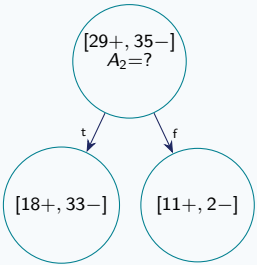
**Open question**

*What happens if features are not binary? What about regression?*

Many frameworks exist. We use Entropy Gain.

Which split is better — $A_1$ or $A_2$?




**Observation**

$A_1$ produces **purer** child nodes ([21+, 5−] and [8+, 30−]) than $A_2$ ([18+, 33−] and [11+, 2−]).

$\Rightarrow$ $A_1$ has higher Information Gain.

Let $p_\oplus$ = proportion of positive examples, $p_\ominus$ = proportion of negative examples in $S$.

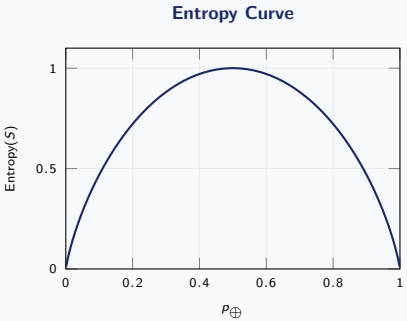
**Binary Entropy**

$$\text{Entropy}(S) \equiv -p_\oplus \log_2 p_\oplus - p_\ominus \log_2 p_\ominus$$

**General Entropy ($C$ classes)**

$$H(S) = -\sum_{c \in C} p_c \log_2 p_c$$

- $H = 0$: set is pure
- $H = 1$: classes equally mixed




Dominant class ⇒ low entropy
Equiprobable ⇒ high entropy

**Information Gain**

$$\text{Gain}(S, A) \equiv \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} \text{Entropy}(S_v)$$

- $\text{Gain}(S, A)$ = expected reduction in entropy due to sorting on $A$
- $S_v$ = subset of $S$ where $A = v$
- Choose the attribute with the **highest** gain

**Equivalently ($IG$ notation)**

$$IG(S, F) = H(S) - \sum_{f \in F} \frac{|S_f|}{|S|} H(S_f)$$

$IG$ measures the **increase in certainty** about $S$ once $F$ is known.

Root node: $S = [9+, 5-]$, $H(S) = -\frac{9}{14}\log_2\frac{9}{14} - \frac{5}{14}\log_2\frac{5}{14} \approx 0.940$

**Feature: Wind**

$$S_{\text{weak}} = [6+, 2-] \;\Rightarrow\; H = 0.811$$
$$S_{\text{strong}} = [3+, 3-] \;\Rightarrow\; H = 1.000$$

$$IG(S, \text{wind}) = 0.940 - \tfrac{8}{14}(0.811) - \tfrac{6}{14}(1.000) = 0.048$$

**All information gains at root:**

| **Attribute** | $IG$ |
|---|---|
| **Outlook** | **0.246** ← maximum |
| Humidity | 0.151 |
| Wind | 0.048 |
| Temperature | 0.029 |

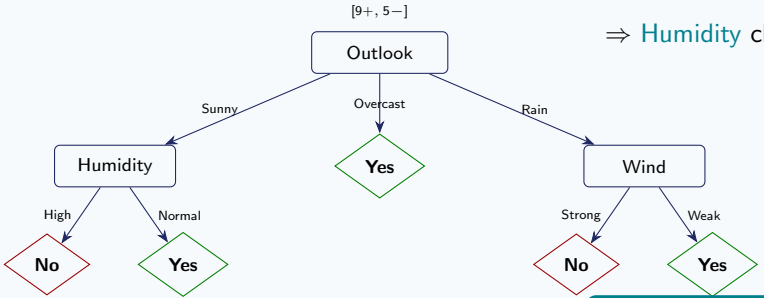
**Outlook** chosen as the root node.

Click here for CAR EXAMPLE

**Sunny branch** ($S_{\text{Sunny}} = \{D1, D2, D8, D9, D11\}$):

$$IG(\text{Humidity}) = 0.970$$
$$IG(\text{Temperature}) = 0.570$$
$$IG(\text{Wind}) = 0.019$$

$\Rightarrow$ Humidity chosen for Sunny branch.




**Algorithm terminates when**

All examples in a node have the **same class label**.

# The Confusion Matrix (Binary)

| | | Predicted: Positive | Predicted: Negative |
|---|---|---|---|
| Actual | Pos | TP | FN |
| | Neg | FP | TN |

- **TP** — True Positive: correctly predicted positive
- **TN** — True Negative: correctly predicted negative
- **FP** — False Positive: negative predicted as positive
- **FN** — False Negative: positive predicted as negative

**Example: Spam Detection**

100 emails: 40 spam, 60 not spam

| | Pred Spam | Pred Ham |
|---|---|---|
| Spam | 35 | 5 |
| Ham | 8 | 52 |

TP=35, FN=5, FP=8, TN=52

# Precision and Recall

## Precision (Positive Predictive Value)

$$\text{Precision} = \frac{TP}{TP + FP}$$

*"Of all predicted positives, how many are actually positive?"*

**Penalizes false alarms (FP)**

## Recall (Sensitivity / True Positive Rate)

$$\text{Recall} = \frac{TP}{TP + FN}$$

*"Of all actual positives, how many did we catch?"*

**Penalizes missed cases (FN)**

## Spam Example (TP=35, FP=8, FN=5)

$$\text{Precision} = \frac{35}{35 + 8} \approx \mathbf{81.4\%}$$

$$\text{Recall} = \frac{35}{35 + 5} = \mathbf{87.5\%}$$

## The Precision-Recall Trade-off

- High threshold $\Rightarrow$ High Precision, Low Recall
- Low threshold $\Rightarrow$ Low Precision, High Recall
- **Cancer detection**: prioritize Recall
- **Spam filter**: prioritize Precision

# F1 Score, $F_\beta$ Score & Specificity

## F1 Score — Harmonic Mean of P & R

$$F_1 = \frac{2\,TP}{2\,TP + FP + FN}$$

Balances Precision and Recall; useful for imbalanced data. Range: $[0, 1]$.

## $F_\beta$ Score (Generalized)

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{Prec} \times \text{Rec}}{\beta^2 \cdot \text{Prec} + \text{Rec}}$$

$\beta > 1$: emphasizes Recall; $\beta < 1$: emphasizes Precision

*E.g., $F_2$ for disease detection, $F_{0.5}$ for spam.*

## Specificity (True Negative Rate)

$$\text{Specificity} = \frac{TN}{TN + FP}$$

"Of all negatives, how many were correctly identified?"

## Spam Example

TP=35, FP=8, FN=5, TN=52

$$F_1 = 2 \cdot \frac{0.814 \times 0.875}{0.814 + 0.875} \approx \mathbf{84.3\%}$$

$$\text{Specificity} = \frac{52}{52 + 8} \approx \mathbf{86.7\%}$$

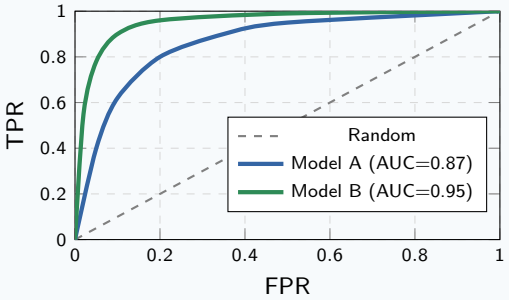
## Key Insight

A well-calibrated model achieves high **Recall** (catches positives) **and** high **Specificity** (avoids false alarms). These two quantities capture complementary aspects of classifier quality.

# ROC Curve & AUC

**ROC** = Receiver Operating Characteristic.
Plots **TPR vs FPR** at every decision threshold.

$$\text{TPR} = \frac{TP}{TP + FN} \qquad \text{FPR} = \frac{FP}{FP + TN}$$




## AUC — Area Under ROC Curve

- **AUC = 1.0**: Perfect classifier
- **AUC = 0.5**: Random guessing
- **AUC < 0.5**: Worse than random

## Interpretation

AUC = probability a random **positive** is ranked higher than a random **negative** by the model.

## When to use AUC

- Comparing classifiers at all thresholds
- Threshold is not yet fixed
- Classes moderately balanced

# Extending to Multiclass: The Confusion Matrix

For $K$ classes, the matrix is $K \times K$:

| | Cat | Dog | Bird |
|---|---|---|---|
| **Cat** | 50 | 5 | 2 |
| **Dog** | 3 | 45 | 4 |
| **Bird** | 1 | 6 | 40 |

**Per-class metrics** (one-vs-rest):

$$\text{Prec}_i = \frac{C_{ii}}{\sum_j C_{ji}}, \quad \text{Rec}_i = \frac{C_{ii}}{\sum_j C_{ij}}$$

**Macro Averaging**

Average **equally** across classes:
Macro-F1 $= \frac{1}{K} \sum_{i=1}^{K} F1_i$
Best when all classes are equally important.

**Weighted Averaging**

Weighted by **class support**: W-F1 $= \frac{\sum_i n_i \cdot F1_i}{\sum_i n_i}$
Good for imbalanced multiclass problems.

**Micro Averaging**

Aggregate TP, FP, FN **globally**:
Micro-F1 $= \frac{2 \sum TP_i}{2 \sum TP_i + \sum FP_i + \sum FN_i}$
Dominated by largest class.

# Macro vs Micro vs Weighted — When to Use What

| Strategy | Best Used When | Watch Out For |
|---|---|---|
| **Macro** | All classes equally important; detecting rare classes matters | Dominated by small classes |
| **Micro** | Large datasets; overall performance | Dominated by large classes |
| **Weighted** | Imbalanced classes; need a fair single metric | Hides poor minority performance |

**Example (Animal Classifier)**

Classes: Cat (57), Dog (52), Bird (47)

$F1_{\text{Cat}} = 0.89, \quad F1_{\text{Dog}} = 0.84, \quad F1_{\text{Bird}} = 0.82$

Macro-F1 $= \frac{0.89+0.84+0.82}{3} = \mathbf{0.85}$

**Key Takeaway**

**Medical diagnosis** with rare classes $\Rightarrow$ **Macro**

**Product categorization** with millions of examples $\Rightarrow$ **Micro or Weighted**

# Cohen's Kappa & Matthews Correlation Coefficient

## Cohen's Kappa ($\kappa$)

Measures agreement adjusted for chance:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

$p_o$ = observed accuracy, $p_e$ = expected by chance

| $\kappa$ | Interpretation |
|---|---|
| $< 0$ | Less than chance |
| 0.0–0.2 | Slight |
| 0.2–0.4 | Fair |
| 0.4–0.6 | Moderate |
| 0.6–0.8 | Substantial |
| 0.8–1.0 | Almost perfect |

## Matthews Correlation Coefficient (MCC)

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$$

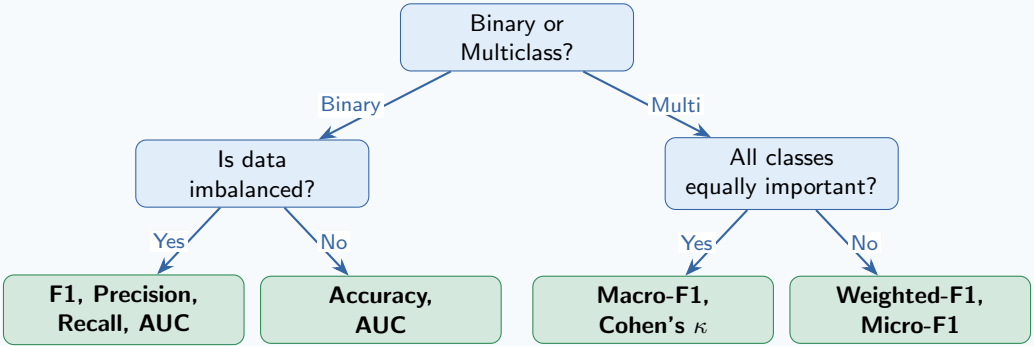
- Range: $[-1, +1]$
- $+1$: Perfect; 0: Random; $-1$: Inverse
- Works well for imbalanced classes
- One of the most informative single metrics

## MCC Example

TP=35, TN=52, FP=8, FN=5

$$\text{MCC} = \frac{35 \cdot 52 - 8 \cdot 5}{\sqrt{43 \cdot 40 \cdot 60 \cdot 57}} \approx \mathbf{0.73}$$

# Decision Framework: Choosing the Right Metric




Always examine the **full confusion matrix** — no single metric tells the whole story.

# Worked Example 1: Medical Diagnosis

## Problem Setup

HIV Test: 200 patients — 30 HIV+, 170 HIV−. Model predicts: 25 HIV+, 175 HIV−

**Confusion Matrix:**

| | **Pred+** | **Pred−** |
|---|---|---|
| **Act+** | 22 | 8 |
| **Act−** | 3 | 167 |

**Metrics:**

$$\text{Accuracy} = \frac{22 + 167}{200} = 94.5\%$$

$$\text{Precision} = \frac{22}{22 + 3} = 88.0\%$$

$$\text{Recall} = \frac{22}{22 + 8} = 73.3\%$$

$$F_1 = \frac{2 \times 0.88 \times 0.733}{0.88 + 0.733} \approx 80.0\%$$

## Analysis

Recall = **73.3%**: 8 HIV+ patients missed. Dangerous in medicine. **Lower the threshold** to catch more positives (increases FP, reduces FN).

# Worked Example 2: Sentiment Analysis (Multiclass)

**Problem: 3-class sentiment — Positive, Neutral, Negative (300 samples each)**

**Per-class results:**

| Class | Prec | Rec | F1 |
|---|---|---|---|
| Positive | 0.82 | 0.88 | 0.85 |
| Neutral | 0.74 | 0.70 | 0.72 |
| Negative | 0.90 | 0.85 | 0.87 |
| **Macro** | 0.82 | 0.81 | **0.81** |
| **Weighted** | 0.82 | 0.81 | **0.81** |

**Observations:**

- Neutral class has lowest F1 (0.72) — harder to classify
- Macro = Weighted here (balanced 300 each)
- When classes are **unbalanced**, Macro $\neq$ Weighted

**Action:** Investigate Neutral class. Consider:

- More training data for Neutral
- Better features for ambiguous sentiment
- Per-class threshold tuning

# Practice Problem 1

## Problem

A fraud detection model is tested on 1000 transactions: 50 fraudulent, 950 legitimate.

| | **Pred Fraud** | **Pred Legit** |
|---|---|---|
| **Fraud** | 40 | 10 |
| **Legit** | 20 | 930 |

Compute: **(a)** Accuracy **(b)** Precision **(c)** Recall **(d)** F1 **(e)** Specificity

## Discussion Question

The bank wants to **minimize fraud losses** (missing fraud is costly). Which metric should they optimize, and should they raise or lower the classification threshold?

# Practice Problem 1 — Solution

$\text{TP} = 40,\ \text{FP} = 20,\ \text{FN} = 10,\ \text{TN} = 930$

$$\text{(a) Accuracy} = \frac{40 + 930}{1000} = \mathbf{97.0}\%$$

$$\text{(b) Precision} = \frac{40}{40 + 20} = \mathbf{66.7}\%$$

$$\text{(c) Recall} = \frac{40}{40 + 10} = \mathbf{80.0}\%$$

$$\text{(d) } F_1 = \frac{2 \times 0.667 \times 0.8}{0.667 + 0.8} \approx \mathbf{72.7}\%$$

$$\text{(e) Specificity} = \frac{930}{930 + 20} = \mathbf{97.9}\%$$

**Discussion Answer**

- Optimize **Recall** — minimize missed fraud (FN)
- **Lower** the threshold $\Rightarrow$ classify more as fraud $\Rightarrow$ fewer missed cases
- Trade-off: more false alarms (FP), lower Precision
- Use $F_2$ **score** ($\beta = 2$) to weight Recall twice as much as Precision

# Practice Problem 2

**Problem: 4-Class Image Classifier**

A model classifies 400 images across 4 classes (100 each):

| | **Cat** | **Dog** | **Car** | **Tree** |
|---|---|---|---|---|
| **Cat** | 85 | 8 | 2 | 5 |
| **Dog** | 6 | 80 | 4 | 10 |
| **Car** | 1 | 2 | 90 | 7 |
| **Tree** | 3 | 5 | 2 | 90 |

**(a)** Compute per-class Precision and Recall for "Dog."

**(b)** Compute overall Accuracy.

**(c)** Which class performs worst? Why?

**(d)** Would you report Macro-F1 or Weighted-F1 here, and why?

# Practice Problem 2 — Solution

**(a) Dog class:**

$$\text{Prec}_{\text{Dog}} = \frac{80}{8+80+2+5} = \frac{80}{95} \approx \mathbf{84.2\%}$$

$$\text{Rec}_{\text{Dog}} = \frac{80}{6+80+4+10} = \frac{80}{100} = \mathbf{80.0\%}$$

**(b) Overall Accuracy:**

$$\text{Acc} = \frac{85+80+90+90}{400} = \frac{345}{400} = \mathbf{86.25\%}$$

**(c) Worst class:**
Dog has the most **false negatives** (20 missed) — particularly confused with Tree (10 cases) and Cat (6 cases).

**(d) Averaging strategy:**
Since all classes have **equal support** (100 each), **Macro-F1 = Weighted-F1**.
To emphasize rare class performance, report Macro-F1.

**Insight**

Even with 86% accuracy, Dog needs improvement — the confusion matrix reveals **where** errors occur.

# Summary & Key Takeaways

## Binary Classification

- **Confusion Matrix** — foundation of all metrics
- **Precision** — quality of predicted positives
- **Recall** — coverage of actual positives
- **F1** — harmonic mean of P & R
- **AUC-ROC** — threshold-free discrimination
- **MCC** — robust even with class imbalance

## Multiclass Classification

- Extend via **one-vs-rest** per class
- **Macro** — equal weight per class
- **Weighted** — proportional to class size
- **Micro** — aggregate all errors globally
- **Cohen's** $\kappa$ — beyond-chance agreement

## Golden Rules

1. **Never** use accuracy alone on imbalanced data
2. Always **inspect** the full confusion matrix
3. Match metric to the **cost of each error type**
4. **No single number** captures full performance
5. Always report **multiple metrics** together

## Key Formulas

$$F_1 = \frac{2\,TP}{2\,TP + FP + FN}$$

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$$